

A COUNTEREXAMPLE TO A LOCAL TOR-LENGTH CONJECTURE OF BOIX AND LIMA-PEREIRA

ABSTRACT. We exhibit complete intersection ideals I, J in a regular local ring of dimension four such that

$$\begin{aligned}\ell(R/(I+J)) &= 8, \\ (\ell(\operatorname{Tor}_i^R(R/I, R/J)))_{i=0}^3 &= (8, 22, 22, 8).\end{aligned}$$

The example disproves Conjecture 4.2 of Boix and Lima-Pereira and gives negative answers to their Questions 5.1 and 5.4.

1. THE COUNTEREXAMPLE

Write ℓ for module length. Boix and Lima-Pereira conjectured in [1, Conjecture 4.2] that if $(R, \mathfrak{m})$ is a regular local ring of dimension n , I is generated by a regular sequence of length $k \leq n-1$, and J is generated by a regular sequence of length n , then

$$(1) \quad \ell(\operatorname{Tor}_i^R(R/I, R/J)) = \binom{k}{i} \ell(R/(I+J)) \quad (0 \leq i \leq k).$$

They also asked whether the corresponding degree-wise lower bound holds when I and J merely contain regular sequences of lengths k and n [1, Question 5.1], and whether

$$(2) \quad \sum_{i=0}^n \ell(\operatorname{Tor}_i^R(R/I, R/J)) \geq 2^k \ell(R/(I+J))$$

holds under the same hypotheses [1, Question 5.4].

Theorem 1. *Let*

$$S = \mathbb{Q}[x, y, z, w], \quad \mathfrak{m} = (x, y, z, w), \quad R = S_{\mathfrak{m}}.$$

Define ideals of S by

$$\begin{aligned}I_S &= (x^2 + xw, y^2 + yw, z^2 + zw), \\ J_S &= (x^2 + xy, xz + z^2, y^2 + yz, xw + w^2),\end{aligned}$$

and set $I = I_S R$ and $J = J_S R$. Then the displayed generators of I and J are regular sequences of lengths 3 and 4, respectively, and

$$\ell(R/(I+J)) = 8.$$

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Moreover,

$$(\ell(\mathrm{Tor}_i^R(R/I, R/J)))_{i=0}^3 = (8, 22, 22, 8),$$

and $\mathrm{Tor}_i^R(R/I, R/J) = 0$ for $i \geq 4$.

Proof. Successively quotienting by the three monic polynomials $x^2 + xw$, $y^2 + yw$, and $z^2 + zw$ shows that they form an S -regular sequence. Moreover, S/I_S is free over $\mathbb{Q}[w]$ with basis

$$\{x^a y^b z^c : a, b, c \in \{0, 1\}\}.$$

Hence their images form an R -regular sequence.

We next show that J_S is $\mathfrak{m}$ -primary. Let P be a prime ideal containing J_S . If $x \in P$, then the second, third, and fourth generators imply $z, y, w \in P$, respectively, so $P = \mathfrak{m}$. If $x \notin P$, then $x + y \in P$. Thus $y \notin P$, and $y + z \in P$. It follows that $z - x \in P$ and $z \notin P$. The relation $z(x + z) \in P$ then gives $x + z \in P$, whence $2x \in P$, a contradiction. Therefore $\sqrt{J_S} = \mathfrak{m}$. The four generators of J form a system of parameters in the Cohen–Macaulay local ring R , and hence an R -regular sequence.

With graded reverse lexicographic order $x > y > z > w$, a Gröbner basis of $I_S + J_S$ is

$$\begin{aligned} & x^2 - w^2, \quad xy + w^2, \quad y^2 + yw, \quad xz - zw, \quad yz - yw, \\ & z^2 + zw, \quad xw + w^2, \quad w^2y, \quad w^2z, \quad w^3. \end{aligned}$$

Its standard monomials are

$$1, \quad x, \quad y, \quad z, \quad w, \quad w^2, \quad yw, \quad zw.$$

Consequently, $I_S + J_S$ is $\mathfrak{m}$ -primary and

$$\ell_S(S/(I_S + J_S)) = 8.$$

A second exact Gröbner-basis computation gives

$$(3) \quad \mathrm{HS}_{(I_S \cap J_S)/(I_S J_S)}(t) = 8t^3 + 14t^4.$$

The computation is reproduced below. Since

$$\mathrm{Tor}_1^S(S/I_S, S/J_S) \cong (I_S \cap J_S)/(I_S J_S),$$

we obtain

$$\ell_S(\mathrm{Tor}_1^S(S/I_S, S/J_S)) = 22.$$

All modules in question are supported at $\mathfrak{m}$, so localization at $\mathfrak{m}$ preserves their lengths and commutes with Tor . Thus

$$\ell_R(\mathrm{Tor}_0^R(R/I, R/J)) = 8, \quad \ell_R(\mathrm{Tor}_1^R(R/I, R/J)) = 22.$$

Koszul duality gives

$$\ell_R(\mathrm{Tor}_i^R(R/I, R/J)) = \ell_R(\mathrm{Tor}_{3-i}^R(R/I, R/J)) \quad (0 \leq i \leq 3);$$

see [1, Proposition 1.9]. Hence the Tor-length vector is $(8, 22, 22, 8)$. Finally, R/I has projective dimension 3, so the higher Tor modules vanish. $\square$

Corollary 2. *Conjecture 4.2 of [1] is false. Questions 5.1 and 5.4 of [1] also have negative answers, even when both ideals are complete intersections.*

Proof. Here $n = 4$, $k = 3$, and $\ell(R/(I + J)) = 8$. For $i = 1, 2$,

$$22 < 24 = \binom{3}{i} 8,$$

which disproves (1) and the lower bound in Question 5.1. Moreover,

$$\sum_{i=0}^4 \ell(\mathrm{Tor}_i^R(R/I, R/J)) = 8 + 22 + 22 + 8 = 60 < 64 = 2^3 \cdot 8,$$

so (2) fails. □

2. EXACT COMPUTATION

The following MACAULAY2 script [2] verifies (3) and the complete Tor-length vector over $\mathbb{Q}$. Since all computed modules have finite length, the command `degree` returns their vector-space dimensions.

```
S = QQ[x,y,z,w, MonomialOrder => GRevLex];

IS = ideal(x^2+x*w, y^2+y*w, z^2+z*w);
JS = ideal(x^2+x*y, x*z+z^2, y^2+y*z, x*w+w^2);

assert(codim IS == 3);
assert(codim JS == 4);
assert(apply(0..4, d -> hilbertFunction(d,S/(IS+JS)))
== {1,4,3,0,0});

Q = (intersect(IS,JS))/(IS*JS);
assert(apply(0..5, d -> hilbertFunction(d,Q))
== {0,0,0,8,14,0});
assert(degree Q == 22);

M = coker gens IS;
N = coker gens JS;
assert(apply(0..4, i -> degree(Tor_i(M,N)))
== {8,22,22,8,0});
```

REFERENCES

- [1] A. F. Boix and B. K. Lima-Pereira, *Exploring a local variant of the Buchsbaum–Eisenbud–Horrocks conjecture*, Proc. Roy. Soc. Edinburgh Sect. A, First View (2026), 1–20; doi:10.1017/prm.2026.10176; arXiv:2607.26118.
- [2] D. R. Grayson and M. E. Stillman, *Macaulay2, a software system for research in algebraic geometry*, available at <https://macaulay2.com/>.